Building and evaluation of a PBPK model for COMPOUND in healthy adults

Version	**master-OSP12.2**
based on *Model Snapshot* and *Evaluation Plan*	https://github.com/Open-Systems-Pharmacology/COMPOUND-Model/releases/tag/vmaster
OSP Version	12.2
Qualification Framework Version	3.5

This evaluation report and the corresponding PK-Sim project file are filed at:

https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/

Table of Contents

1 Intro

COMPOUND is an active, highly selective ... (Information about Pharmacology)

COMPOUND is ... (Information about relevant Pharmacokinetics)

The herein presented model building and evaluation report evaluates the performance of the PBPK model for COMPOUND in (healthy) adults.

The presented COMPOUND PBPK model as well as the respective evaluation plan and evaluation report are provided open-source (https://github.com/Open-Systems-Pharmacology/COMPOUND-Model).

Alfentanil is a potent analgesic synthetic opioid. It is fast but short-acting and used for anesthesia during surgery. Alfentanil is metabolized solely by CYP3A4 (Phimmasone 2001). Like midazolam, alfentanil is not a substrate for P-gp (Wandel 2002) and less than 1% of an alfentanil dose is excreted unchanged in urine (Meuldermans 1988).

Although in clinical use alfentanil is always administered intravenously (iv), some DDI studies published plasma concentration-time profiles of alfentanil following oral ingestion. The presented alfentanil model was established using clinical PK data of 8 publications, covering iv and oral (po) administration and a dosing range from 0.015 to 0.075 mg/kg as well as absolute doses of 1 mg iv and 4 mg po. The established model is based on the model developed by Hanke et al. (Hanke 2018) and applies metabolism by CYP3A4 and glomerular filtration.

2 Methods

2.1 Strategy

2.2 Data

2.3 Assumptions

3 Results

3.1 Parameters

Compound: Dabigatran

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	17 mg/l		Measurement	True
Reference pH	7		Measurement	True
Lipophilicity	1.1436481967 Log Units	Parameter Identification-Parameter Identification-Value updated from '#7 - iv (logP, logP, UGT, GFR)_best iv' on 2025-08-30 15:56	Measurement	True
Fraction unbound (plasma, reference value)	0.65		Measurement	True
Permeability	4.6547688496E-06 dm/min	Parameter Identification-Parameter Identification-Value updated from '#1 update' on 2025-08-14 16:39	FIT	False
Is small molecule	Yes			
Molecular weight	471.51 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Systemic Process: Glomerular Filtration-Dabi

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	

Metabolizing Enzyme: UGT2B15-FIT

Molecule: UGT2B15

Metabolite: DabiGluc

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Km	512 µmol/l	
kcat	4.2993761606 1/min	Parameter Identification-Parameter Identification-Value updated from '#7 - iv (logP, logP, UGT, GFR)_best iv' on 2025-08-30 15:56

3.2 Plots

Table 3-1: GMFE for Goodness of fit plot for concentration in plasma

Group	GMFE
Intravenous infusion	1.16
Oral capsule	1.28
Oral solution	1.92
All	1.32

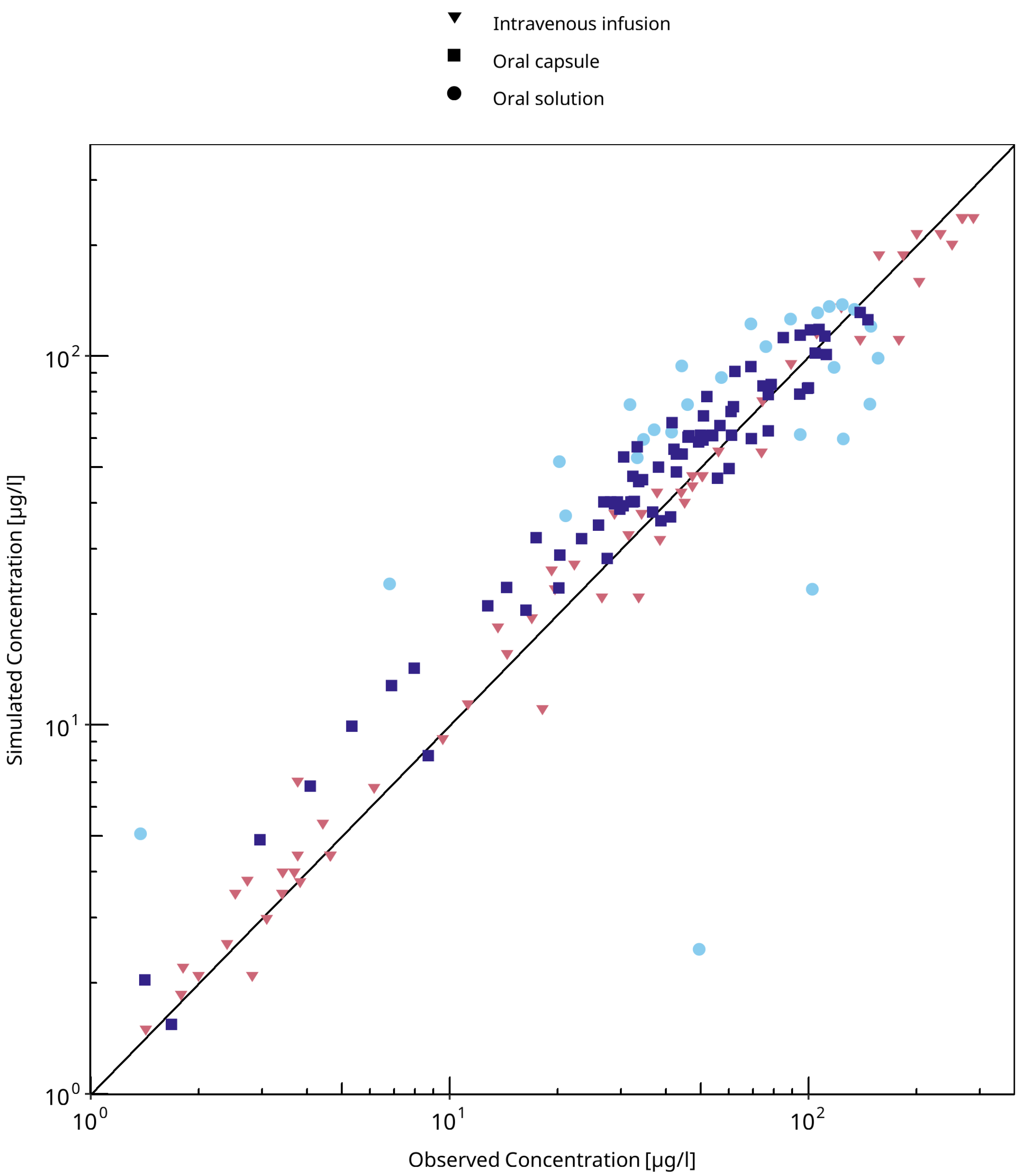

Figure 3-1: Goodness of fit plot for concentration in plasma

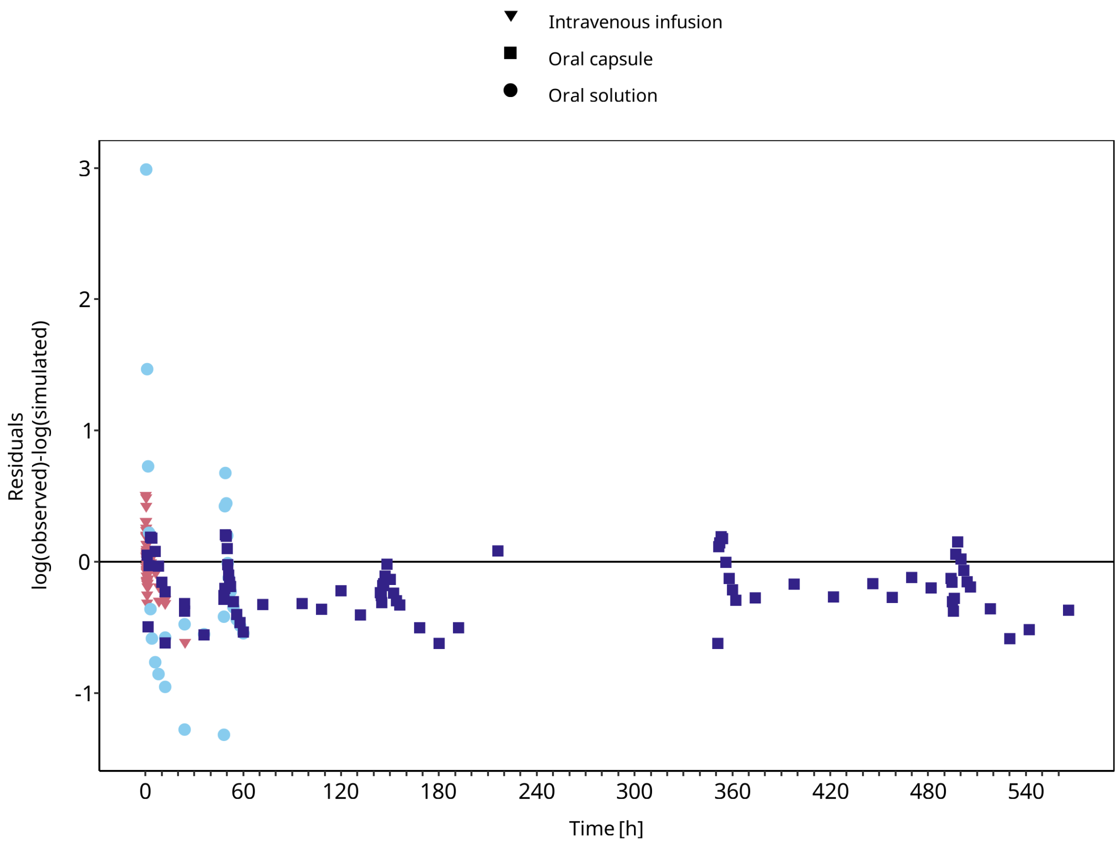

Figure 3-2: Goodness of fit plot for concentration in plasma

3.3 Profiles

3.3.1 Training

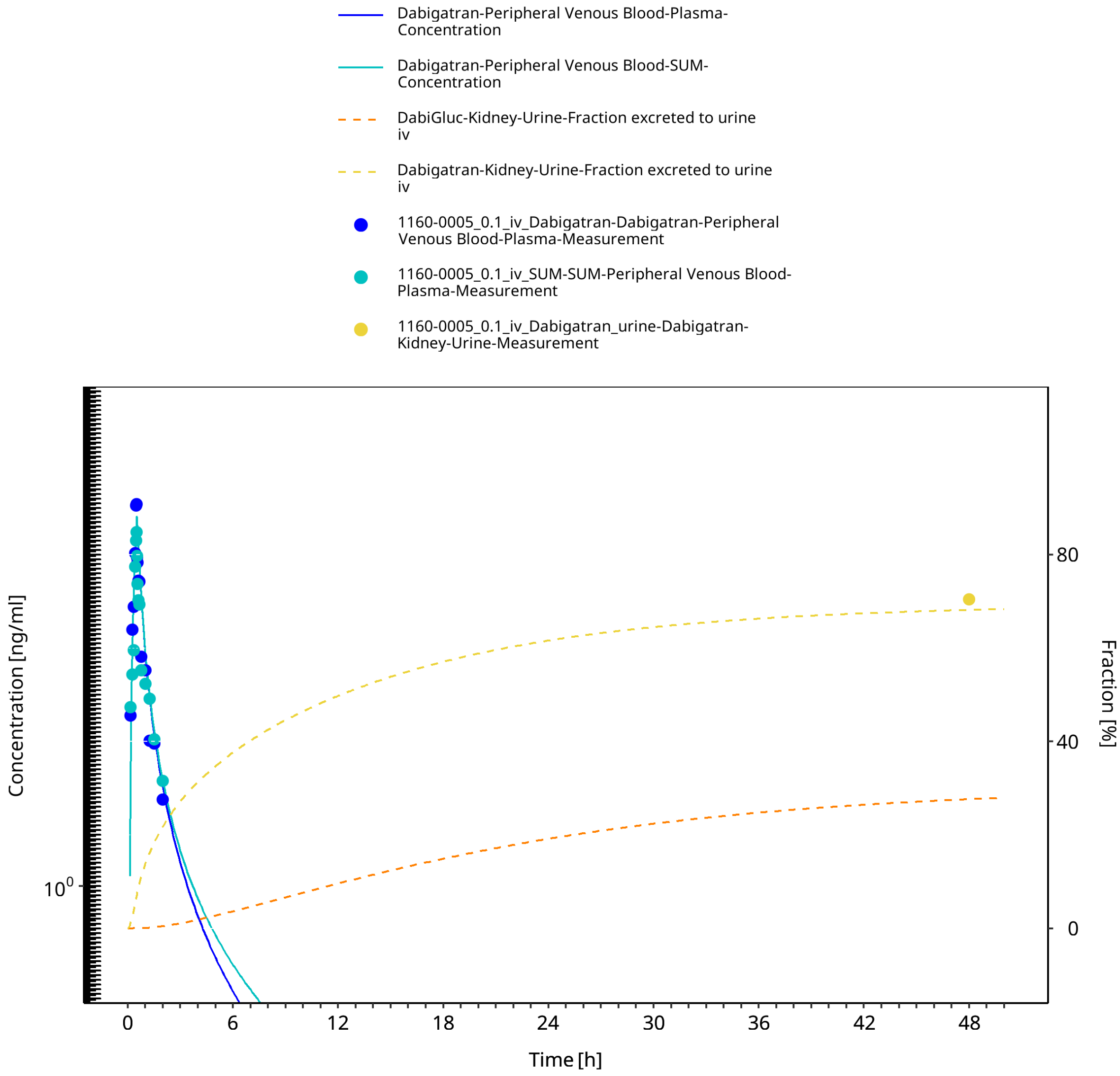

Figure 3-3: 1160-0005 - Dabigatran iv 0.1 mg - log

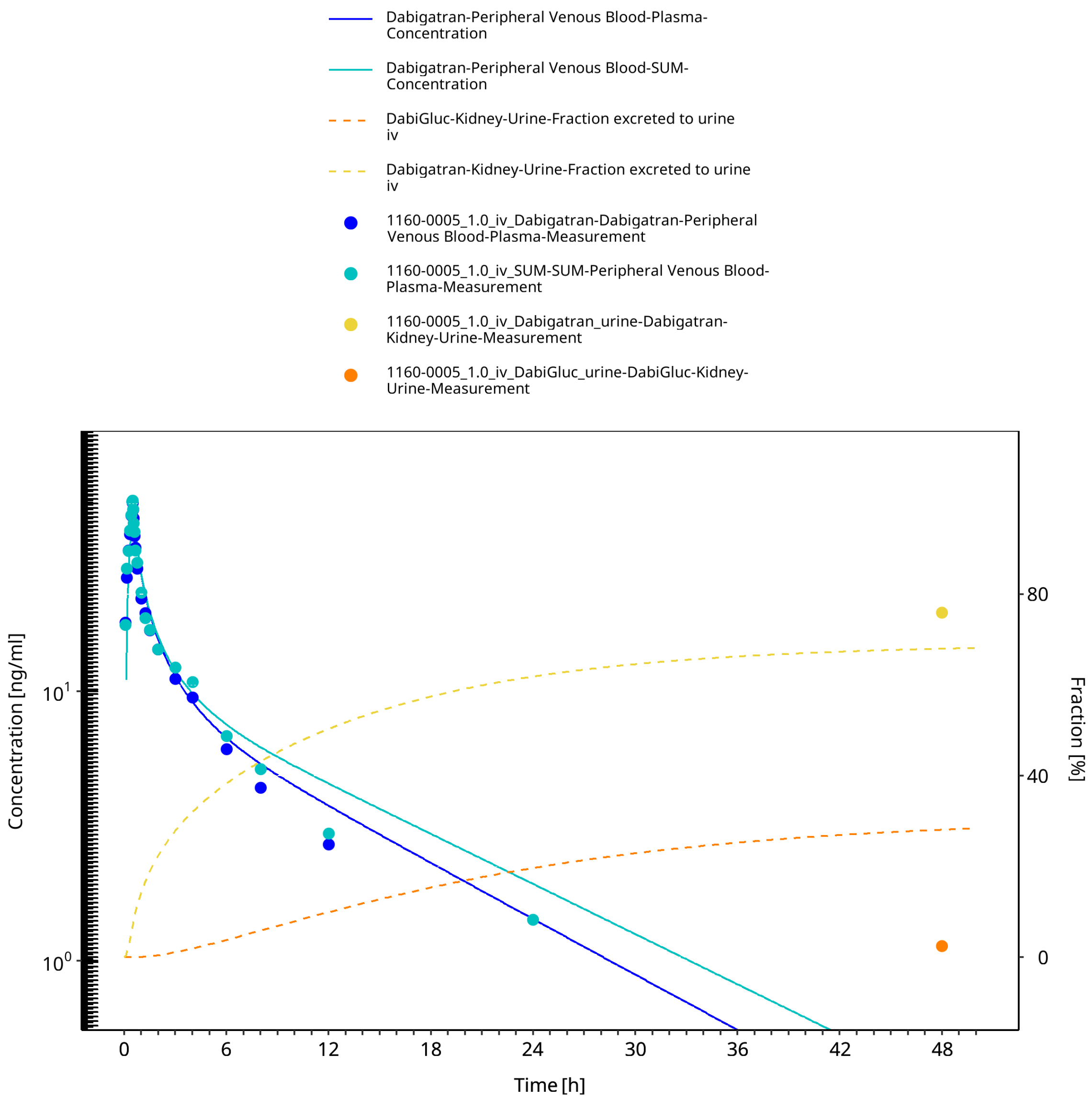

Figure 3-4: 1160-0005 - Dabigatran iv 1.0 mg - log

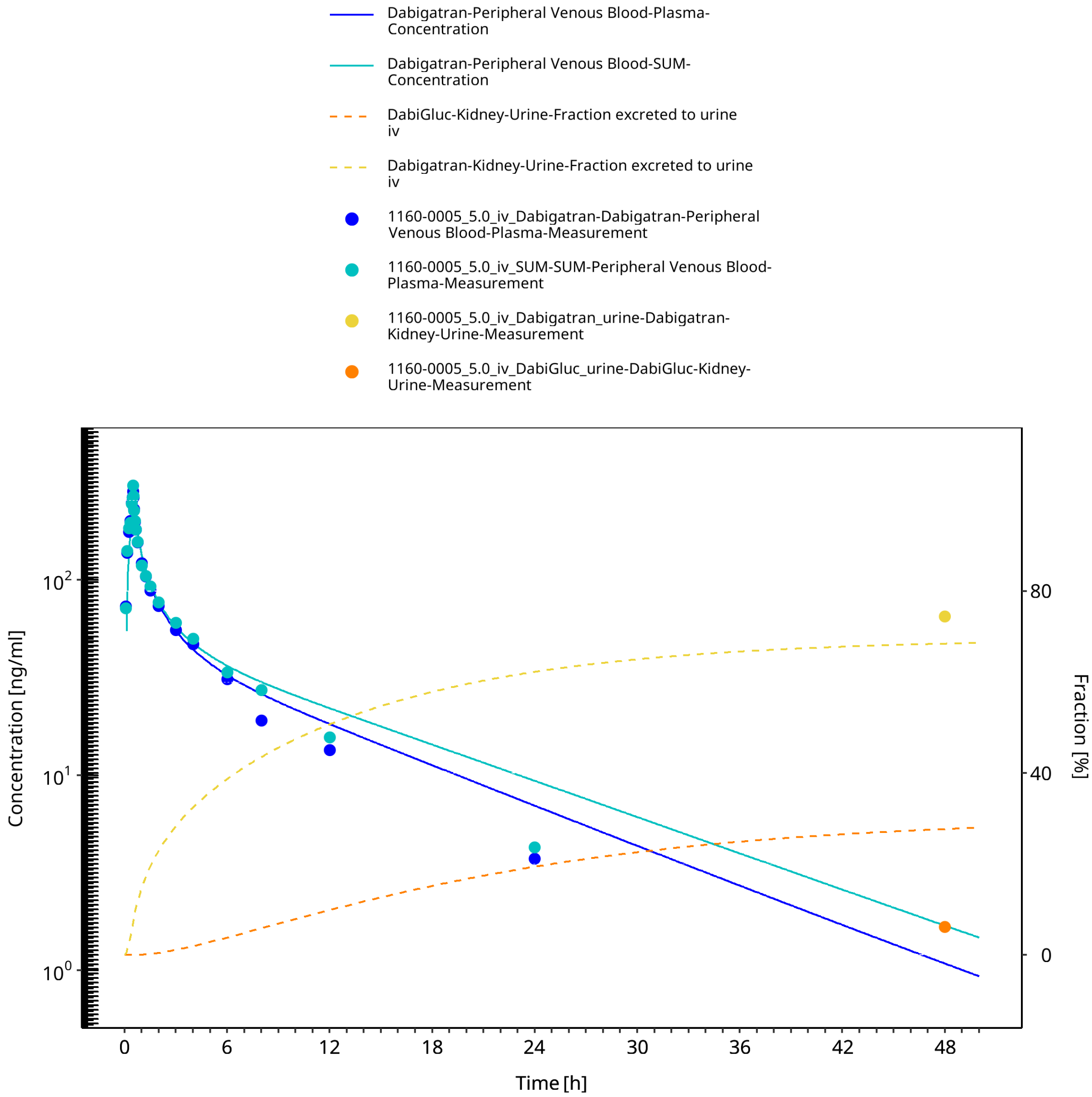

Figure 3-5: 1160-0005 - Dabigatran iv 5.0 mg - log

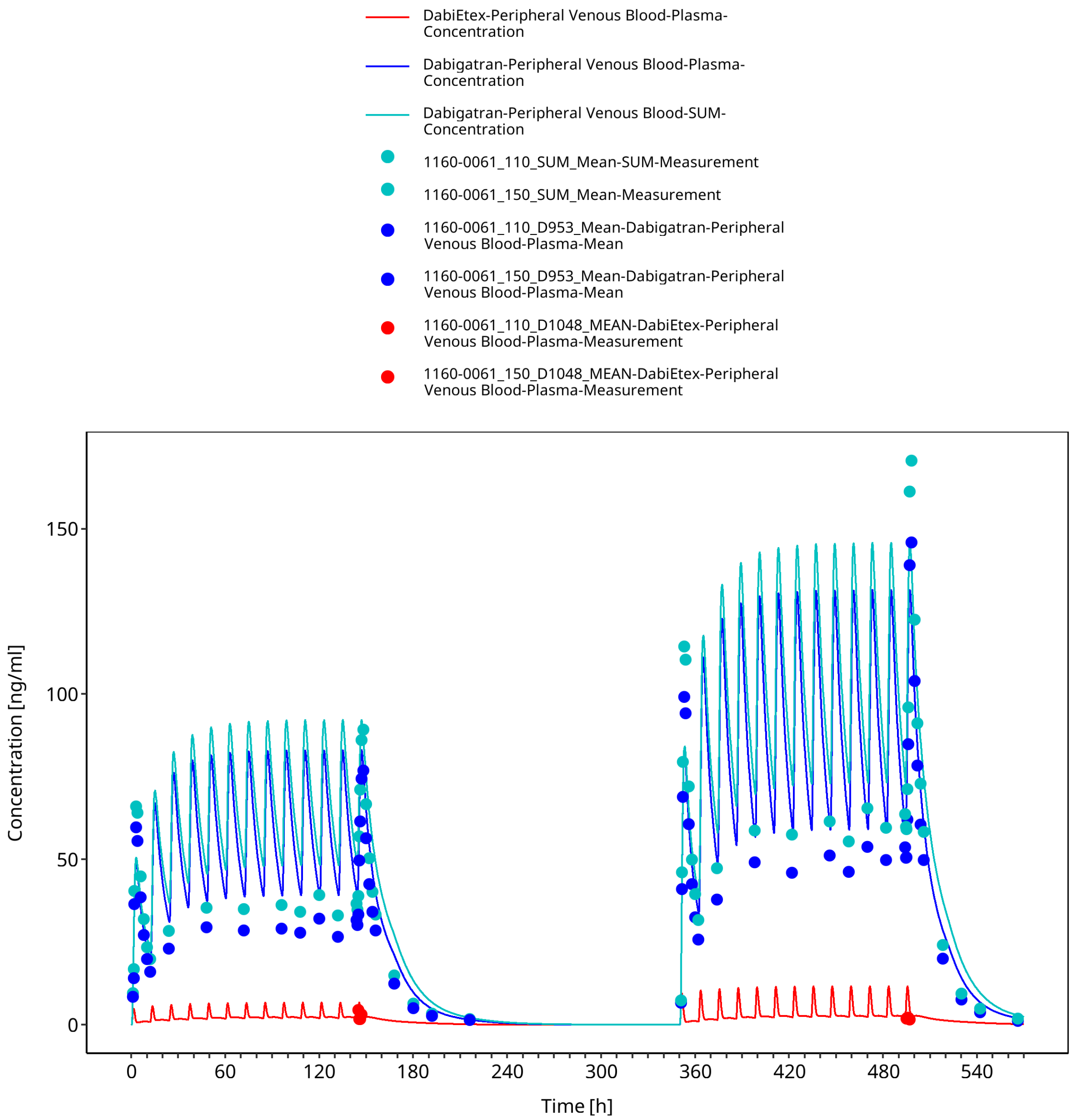

Figure 3-6: 1160-0061 - Dabigatran 110+150 mg BID - lin

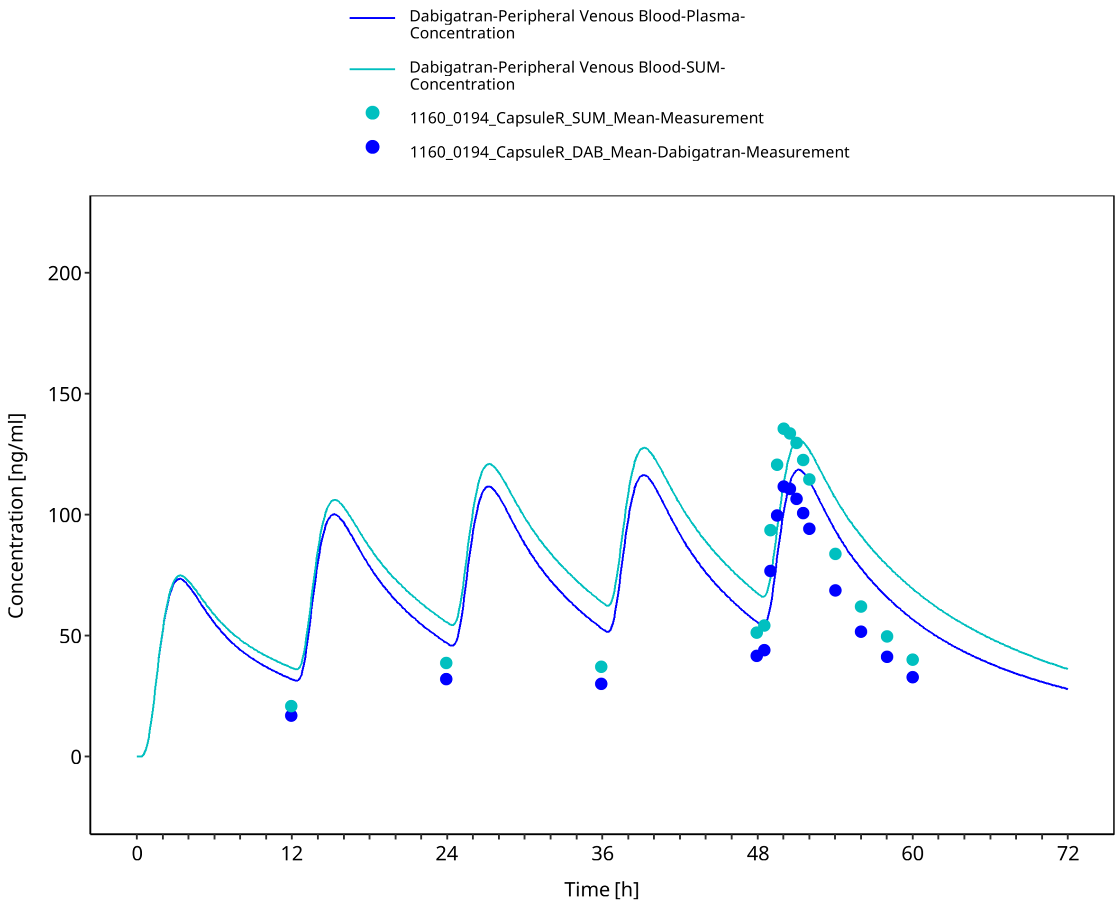

Figure 3-7: 1160-0194 - Dabigatran 150 mg Capsule - lin

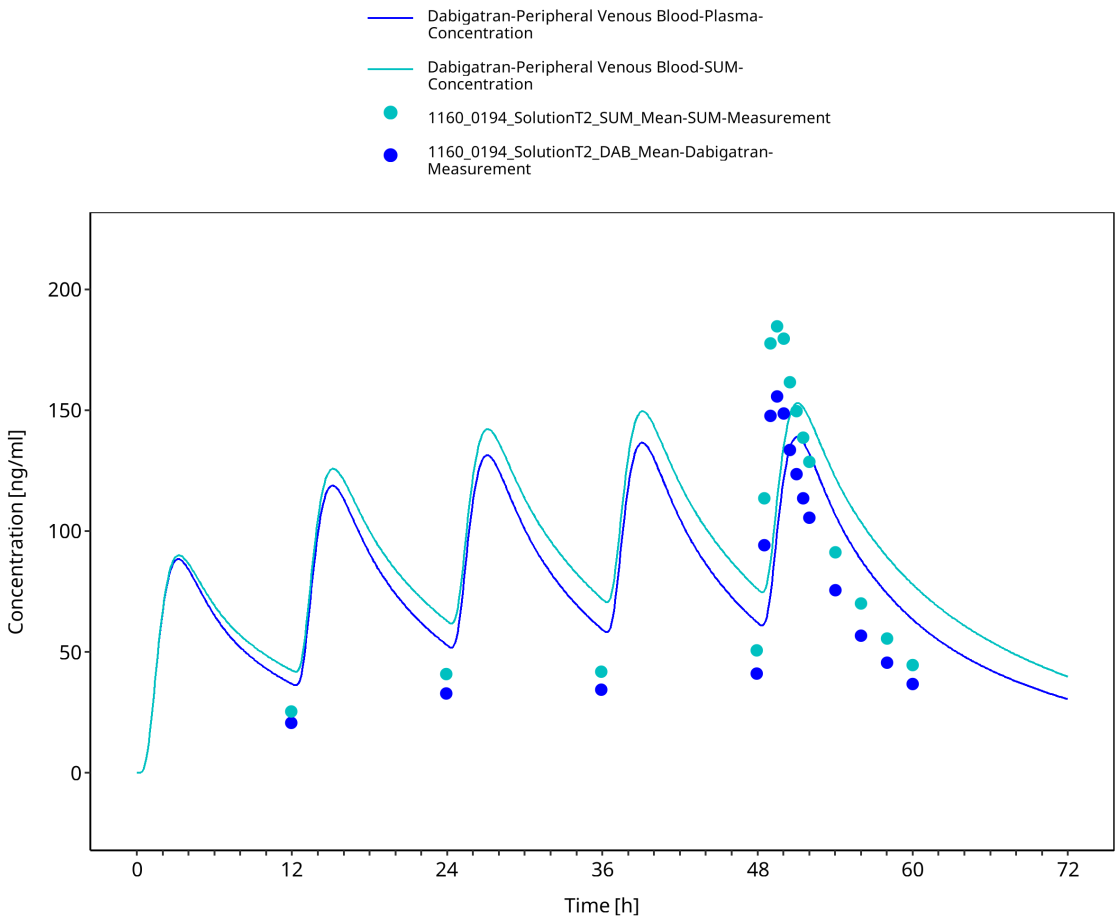

Figure 3-8: 1160-0194 - Dabigatran 150 mg Solution - lin

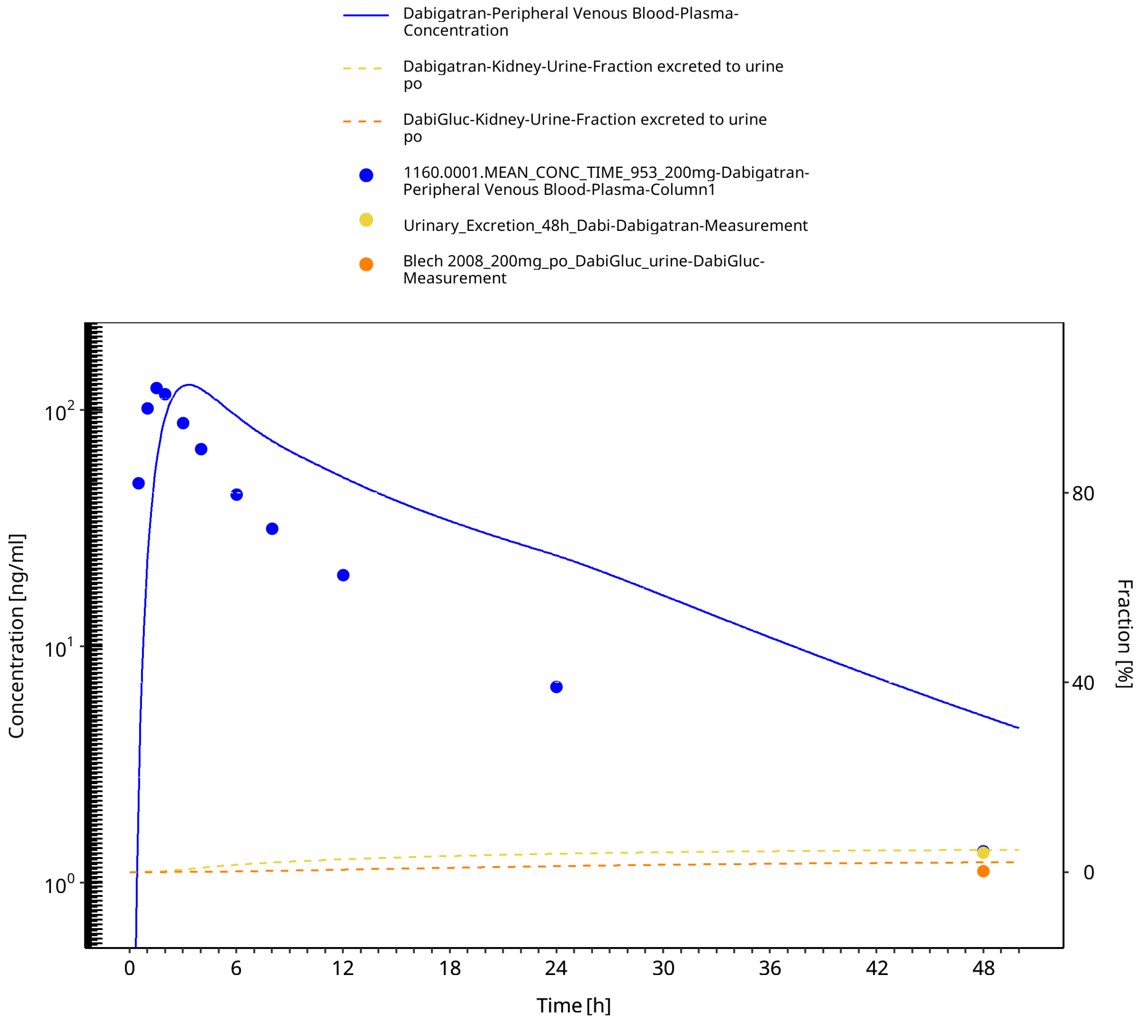

Figure 3-9: Blech 2008 - Dabigatran 200 mg Solution - log

3.3.2 Test

4 Conclusion

5 References